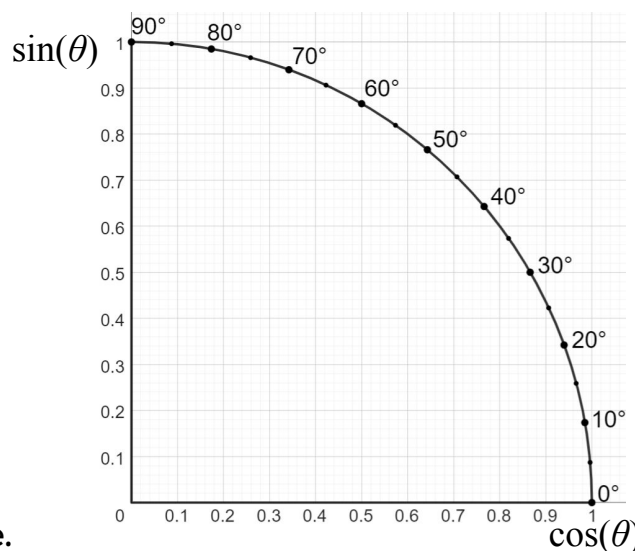




The unit circle

Task A: Introduction to sine and cosine

The diagram shows the sector of the unit circle that lies in the first quadrant.



1) a) Use the diagram to complete the table.

θ°	0	10	20	30	40	50	60	70	80	90
$\sin(\theta)$	0	0.17	0.34	0.5	0.64	0.77	0.87	0.94	0.98	1

Answers may vary slightly depending on how accurately they are read from the graph.

Use the link for an interactive version: [unit circle in first quadrant](#)

b) What happens to the value of $\sin(\theta)$ as the angle increases in this quadrant?

It increases. Its lowest value is 0 and greatest value is 1.

c) Use your table to find the angle where $\sin(\theta) = 0.5$. 30°

d) Use the diagram to find the angle where $\sin(\theta) = 0.8$. 53°

2) a) Use the diagram to complete the table.

θ°	0	10	20	30	40	50	60	70	80	90
$\cos(\theta)$	1	0.98	0.94	0.87	0.77	0.64	0.5	0.34	0.17	0

Answers may vary slightly depending on how accurately they are read from the graph.

Use the link for an interactive version: [unit circle in first quadrant](#)

b) What happens to the value of $\cos(\theta)$ as the angle increases in this quadrant?

It decreases. Its lowest value is 0 and greatest value is 1.

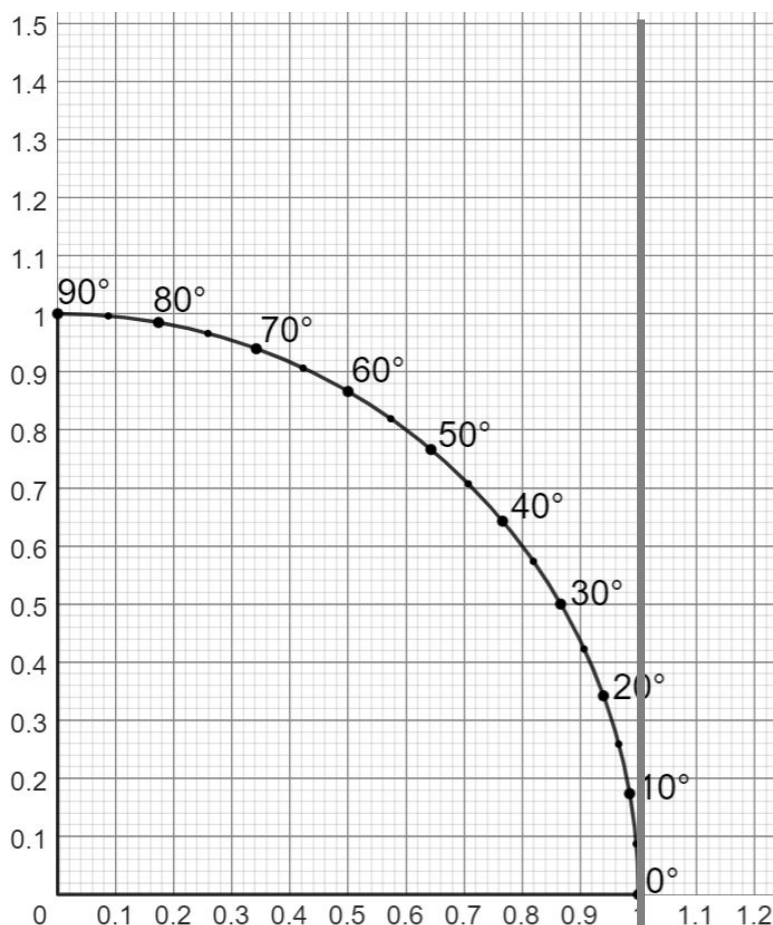
c) Find the angle where $\cos(\theta) = 0.5$. 60°

d) Find the value of θ where $\cos(\theta) = \sin(40^\circ)$. 50°

e) Find the angle where $\cos(\theta) = \sin(\theta)$. 45°



Task B: Introduction to tangent



For an interactive version of the diagram, click the link: [unit circle in first quadrant](#)

1) The diagram shows the first quadrant of the unit circle.

a) Draw a tangent at $x = 1$.

b) Find the value of $\tan(35^\circ)$

0.7

c) Find the value of $\tan(55^\circ)$

1.4

d) Use the diagram to find the angle where $\tan(\theta) = 1$.

45°

e) Use the diagram to find the angle where $\tan(\theta) = 0.5$.

27°

Answers may vary slightly depending on how accurately they are read from the graph.

2) a) Complete the table.

θ°	30	40	50
$\sin(\theta)$	0.5	0.64	0.77
$\cos(\theta)$	0.87	0.77	0.64
$\tan(\theta)$	0.58	0.84	1.19

Answers may vary depending on accuracy.

b) For each angle, divide the value of $\sin(\theta)$ by the value of $\cos(\theta)$. What do you notice?

The result of each division is approximately equal to the value of $\tan(\theta)$.

